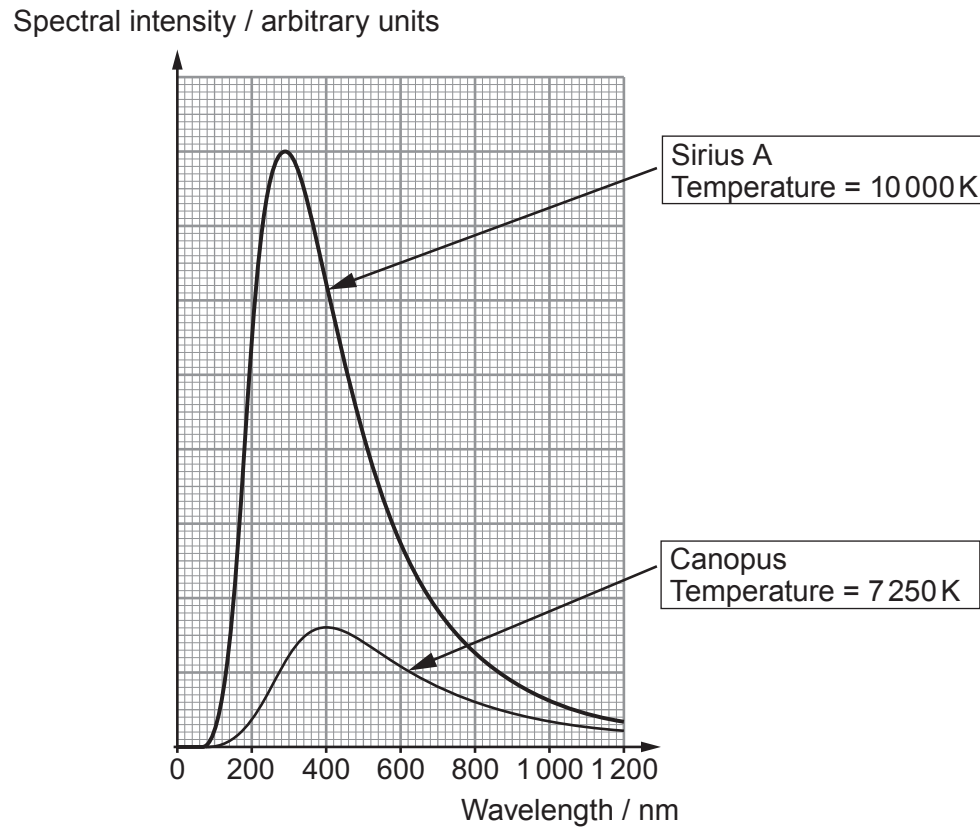


3. (a) Two of the brightest stars in the night sky are Sirius A and Canopus. The graph shows the continuous black body spectra for these two stars.



- (i) Confirm that Wien's displacement law is valid for the stars. [3]

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(ii) Astronomers state that one of the stars appears ‘bluer’ than the other. Explain how the spectra support this statement and state which star would appear bluer. [2]

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(b) (i) The radius of Canopus is 4.97×10^{10} m and that of Sirius A is 1.19×10^9 m. Show that the luminosity of Canopus is approximately 500 times the luminosity of Sirius A. [4]

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(ii) Calculate the intensity of the radiation reaching the surface of the Earth from Sirius A. (Distance between Sirius A and Earth = 8.15×10^{16} m.) [2]

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(iii) The intensity of the radiation reaching the Earth’s surface from Canopus is **less than** that from Sirius A, even though Canopus has a **greater** luminosity than Sirius A. Explain this apparent contradiction. [2]

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